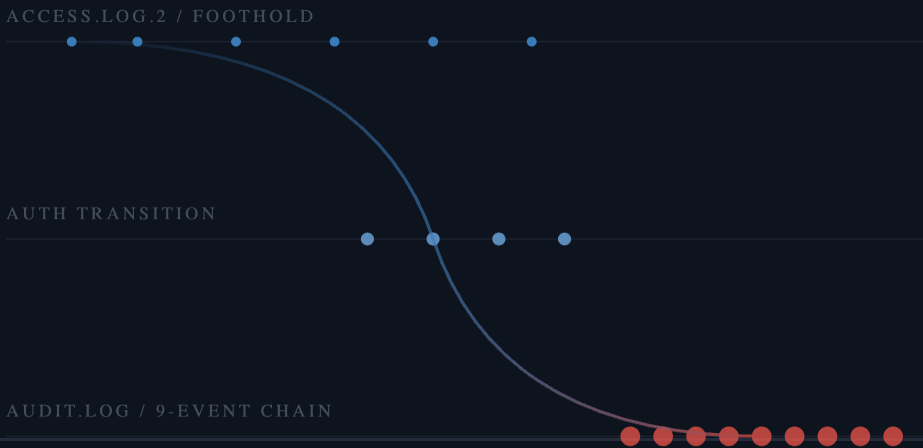


CASE FILE // AIT-LDS V2.0 / RUSSELLMITCHELL

Reconstructing a privilege escalation attack from raw logs.

A 3NF PostgreSQL schema turns an access.log.2 foothold into a 9-event su + sudo chain on intranet_server.



TEAM	INSTRUCTOR	DELIVERABLES
Naman Sudan River Roseveare-Hunt Ishaan Shetty	Dr. Liu	22-table schema 23 SQL queries 9-panel dashboard 10-section report
	STACK	
	PostgreSQL 16 · Alembic SQLAlchemy · Streamlit · Plotly	

How we built it.

GitHub



- Version control with reviewed pull requests.
- Single source of truth for schema, queries, dashboard.

Linear



- Sprint planning against rubric line items.
- Task tracking and ownership.

Slack



- Day-to-day team communications.

Docker



- Reproducible PostgreSQL 16 environment.
- One-command database rebuild from scratch.

Alembic



- Schema migrations versioned in Git.
- Every table change ships as a migration.

Streamlit + Plotly

- Interactive dashboard layer.
- Hits PostgreSQL directly via SQLAlchemy.

STACK · POSTGRESQL 16 + SQLALCHEMY + PYTHON LOADERS FOR THE DATA PLANE.

A 22-table 3NF schema over the russellmitchell testbed.

22

HOSTS IN INVENTORY

8

SOURCE LOG FILES

61,862

LABELED ATTACK LINES

~439K

LOADED ROWS

SOURCE DATA, THREE DOMAINS

Host inventory: 7 tables, including host, fqdn, ip_address, distribution.

Audit events: 10 tables, with audit_event as the supertype and 8 subtype tables for the audit-type families.

Attack labels: 5 tables, including attack_phase, attack_label, labeled_line, plus a per-rule annotation table.

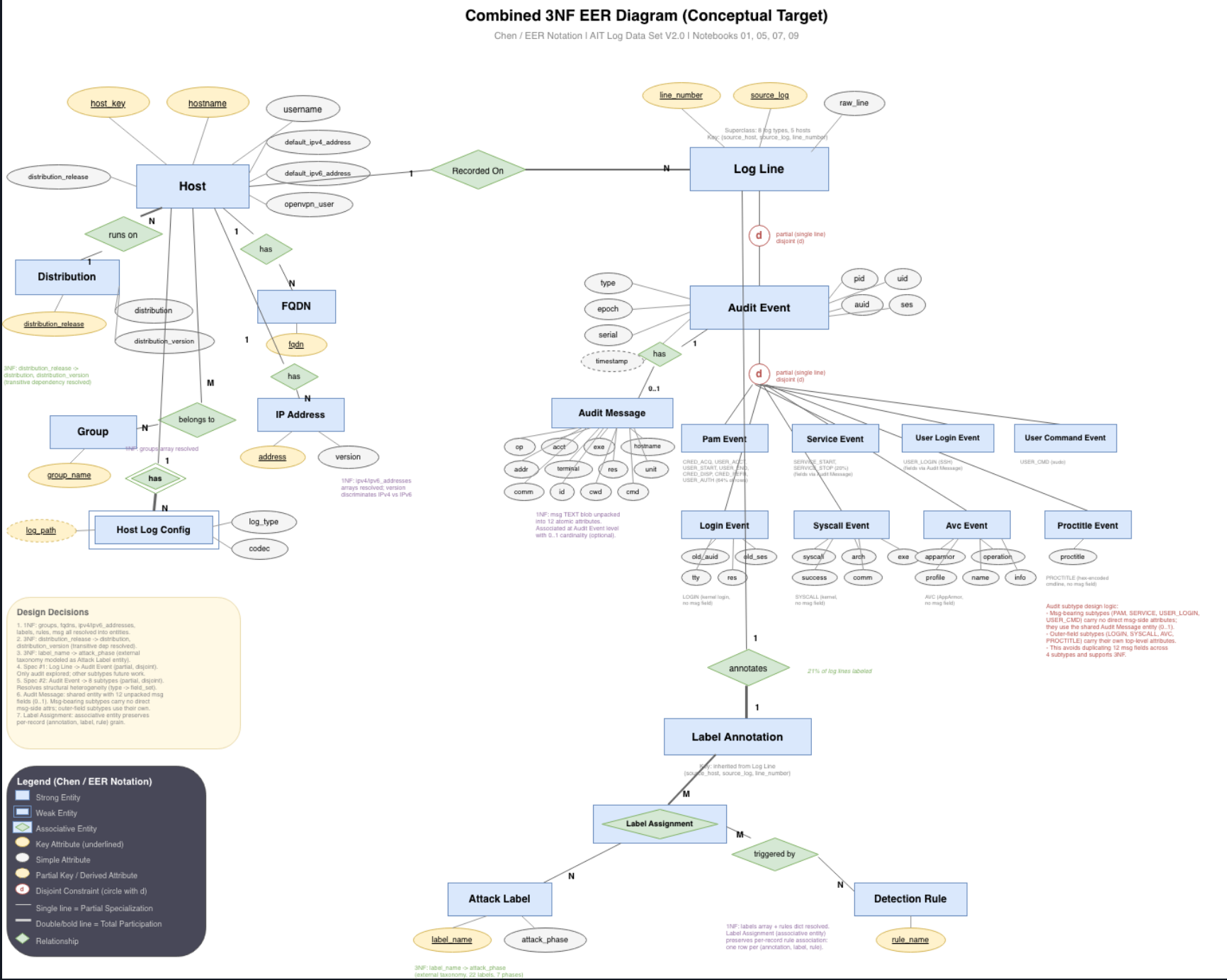
STORY SETUP

An access.log.2 foothold leads into the audit.log su then sudo chain on intranet_server.

TWO LOG FAMILIES. ONE HOST. NINE AUDIT EVENTS.

A 22-table 3NF schema over the russellmitchell testbed.

CONCEPTUAL 3NF SCHEMA · EER DIAGRAM



SCHEMA SUMMARY

22
TABLES

3
DOMAINS

3NF
NORMAL FORM

DOMAINS

Host inventory · 7 tables host, fqdn, ip_address, distribution, ...

Audit events · 10 tables audit_event supertype + 8 subtypes, audit_message

Attack labels · 5 tables attack_phase, attack_label, labeled_line, labeled_line_label, labeled_line_rule

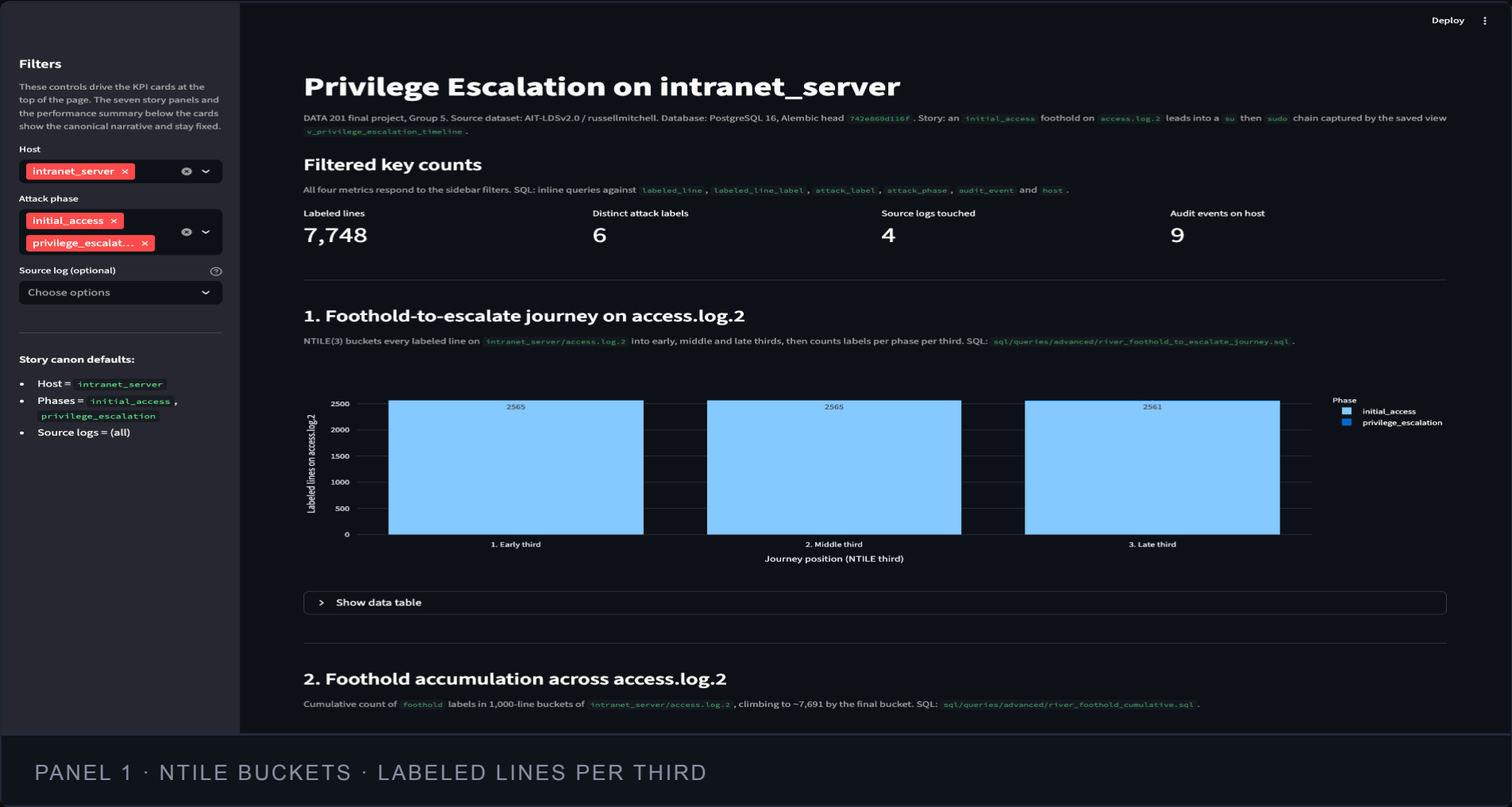
23 queries in the repo. The slides spotlight four.

Full portfolio lives in GitHub and report section 6. The spotlight rows below are the deep dives that drive the next four slides.

AUTHOR	QUERY	TECHNIQUE TAGS
River	Foothold-to-escalation journey river_foothold_to_escalate_journey.sql	CTE NTILE WINDOW
	Foothold accumulation across access.log.2	CUMULATIVE SUM WINDOW
	Source spread per host, host and label baselines	RANK JOIN ≥3 BASIC
Ishaan	Audit types in privilege escalation ishaan_audit_types_in_privilege_escalation.sql	EXISTS SUBQUERY
	Detection rules ranked, top rule per label	RANK WINDOW CORRELATED SUBQUERY
	Phase and rule baseline counts	BASIC
Naman	EXPLAIN ANALYZE + composite index naman_explain_index_improvement.sql	EXPLAIN ANALYZE INDEX
	Saved view v_privilege_escalation_timeline naman_view_privilege_escalation_timeline.sql	SAVED VIEW JOIN ≥8
	Audit-type density per phase, service baselines	DENSE_RANK WINDOW BASIC

Where in access.log.2 does the foothold turn into escalation evidence?

```
WITH access_labels AS (  
  
  SELECT ll.line_number,  
         al.label_name,  
         ap.phase_name,  
  
  NTILE(3) OVER (ORDER BY ll.line_number) AS journey_third  
  
  FROM labeled_line ll  
  
  JOIN labeled_line_label lll ON lll.labeled_line_id = ll.labeled_line_id  
  
  JOIN attack_label al ON al.label_id = lll.label_id  
  
  JOIN attack_phase ap ON ap.phase_id = al.phase_id  
  
  WHERE ll.source_host = 'intranet_server' AND ll.source_log = 'access.log.2' AND ap.phase_name IN  
    ('initial_access', 'privilege_escalation' )  
)  
SELECT journey_third, phase_name, COUNT(*) AS labeled_lines FROM access_labels GROUP BY  
journey_third, phase_name ORDER BY journey_third, phase_name;
```



An 8-table saved view that becomes the privilege-escalation timeline.

```
CREATE OR REPLACE VIEW v_privilege_escalation_timeline AS SELECT h.host_key AS host,
    ae.timestamp,
    ae.type
AS audit_type,
    am.op
AS pam_operation,
    am.acct
AS target_account,
    am.exe
AS executable,
    am.terminal,

string_agg(DISTINCT al.label_name, ',' ORDER BY al.label_name) AS labels,

string_agg(DISTINCT llr.rule_name, ',' ORDER BY llr.rule_name) AS rules FROM audit_event ae JOIN host h ON h.host_id =
ae.host_id LEFT JOIN audit_message am ON am.event_id = ae.event_id JOIN labeled_line ll ON ll.source_host = h.host_key

AND ll.source_log = 'audit.log' AND ll.line_number = ae.line_number JOIN labeled_line_label lll ON lll.labeled_line_id = ll.labeled_line_id
JOIN attack_label al ON al.label_id = lll.label_id JOIN attack_phase ap ON ap.phase_id = al.phase_id LEFT JOIN labeled_line_rule llr ON
llr.labeled_line_id = ll.labeled_line_id

AND llr.label_id = al.label_id WHERE ap.phase_name = 'privilege_escalation' GROUP BY h.host_key, ae.timestamp, ae.type, am.op, am.acct,
am.exe, am.terminal;
```

V_PRIVILEGE_ESCALATION_TIMELINE · INTRANET_SERVER			
TIMESTAMP	AUDIT_TYPE	PAM_OP · ACCT	LABELS
15:53:55	USER_AUTH	PAM:auth · root	su_attempt
15:53:55	USER_ACCT	PAM:acct · root	su_attempt
15:53:55	CRED_ACQ	PAM:setcred · root	su_attempt
15:53:55	USER_START	PAM:session · root	su_session
15:54:11	USER_END	PAM:session · root	su_session
15:54:11	CRED_DISP	PAM:setcred · root	su_session
15:54:23	USER_CMD	cmd · jhall	sudo_exec
15:54:23	CRED_REFR	PAM:setcred · root	sudo_exec
15:54:23	USER_START	PAM:session · root	sudo_session

One view, two bursts. **Burst 1 at 15:53:55 is su; burst 2 at 15:54:23 is sudo.** The dashboard reads from this view, not from a giant ad-hoc join. The SQL is the abstraction.

TABLES	AGGREGATOR	PHASE	ROWS
8	string_agg	privilege_escalation	9

Which audit types fire on intranet_server lines that carry a privilege_escalation label?

```
SELECT DISTINCT ae.type AS audit_type FROM audit_event ae JOIN host h ON h.host_id = ae.host_id WHERE
h.host_key = 'intranet_server' AND EXISTS (

SELECT 1 FROM labeled_line ll

JOIN labeled_line_label ll_label ON ll_label.labeled_line_id = ll.labeled_line_id

JOIN attack_label al ON al.label_id = ll_label.label_id

JOIN attack_phase ap ON ap.phase_id = al.phase_id

WHERE ll.source_host = h.host_key

AND ll.source_log = 'audit.log' AND ll.line_number = ae.line_number

AND ap.phase_name = 'privilege_escalation' ) ORDER BY audit_type;
```

PATTERN	RESULT	RUNTIME
EXISTS · 4-tbl body	8 rows	0.233 s

CRED FAMILY · 3

CRED_ACQ

CRED_DISP

CRED_REFR

USER FAMILY · 5

USER_ACCT

USER_AUTH

USER_CMD

USER_END

USER_START

	audit_type
1	CRED_ACQ
2	CRED_DISP
3	CRED_REFR
4	USER_ACCT
5	USER_AUTH
6	USER_CMD
7	USER_END
8	USER_START

RUN IN PGADMIN 0.233 SEC · 8 ROWS

AUDIT_TYPE, CHARACTER VARYING(20)

The chain triggers a specific subset of the auditd taxonomy, only credential and user-session events plus one USER_CMD. A detector subscribed to those 8 types catches the chain without the full auditd firehose.

Sequential scan to index scan on the incident-response zoom.

Workload: h.host_key = 'intranet_server', ae.timestamp in 2022-01-24 04:37:30 to 04:38:15 UTC. Index: idx_audit_event_host_timestamp ON audit_event (host_id, timestamp), migration 742e860d116f.

• BEFORE · NO INDEX

1.462ms

PLAN NODE · SEQ SCAN ON AUDIT_EVENT

BUFFERS (SHARED HIT)

141

PLANNER COST

171.22

• AFTER · IDX_AUDIT_EVENT_HOST_TIMESTAMP

0.640ms

PLAN NODE · INDEX SCAN USING COMPOSITE INDEX

BUFFERS (SHARED HIT)

33

PLANNER COST

24.79

EXECUTION SPEEDUP

~2.3x

1.462 ms → 0.640 ms

BUFFERS READ

~4x

141 → 33 fewer

PLANNER COST

~7x

171.22 → 24.79

Composite index design must match query shape: equality on host_id first, range on ae.timestamp second.

Streamlit + SQLAlchemy + Plotly, hitting PostgreSQL directly.

Sidebar filters: host (default intranet_server), attack phase (default initial_access + privilege_escalation), source log (multi-select).

• DEFAULT FILTERS · HOST = INTRANET_SERVER

7,748
LABELED LINES

6
DISTINCT LABELS

4
SOURCE LOGS

9
AUDIT EVENTS

• HOST FILTER CLEARED · ALL HOSTS

8,806
LABELED LINES

7
DISTINCT LABELS

7
SOURCE LOGS

9
AUDIT EVENTS

Filters

These controls drive the KPI cards at the top of the page. The seven story panels and the performance summary below the cards show the canonical narrative and stay fixed.

Host: intranet_server

Attack phase: initial_access, privilege_escalation

Source log (optional): Choose options

Story canon defaults: Host = intranet_server, Phases = initial_access, privilege_escalation, Source logs = (all)

2. Foothold accumulation across access.log.2

Cumulative count of foothold labels in 1,000-line buckets of intranet_server/access.log.2, climbing to ~7,693 by the final bucket. SQL: sql/queries/advanced/river_2/foothold_cumulative.sql.

Cumulative footholds

Line number bucket (start of 1,000-line window)

Foothold lines per bucket

3. Privilege escalation timeline (su then sudo)

Nine-event chain on intranet_server between 2022-01-24 04:37:40 and 04:38:06 UTC. Source: Alembic-managed view v_privilege_escalation_timeline, exposed by sql/queries/advanced/main_vie_privilege_escalation_timeline.sql.

Host	Event timestamp	Audit type	PAM op	Target account	Executable	Terminal	Labels	Rules
intranet_server	2022-01-24 04:37:40 UTC	USER_AUTH	PAMauthentication	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_ACCT	PAMaccounting	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	CRED_ACQ	PAMsetcred	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:37:40 UTC	USER_START	PAMsession_open	jhall	/bin/su	/dev/pts/1	attacker_change_user, escalate	attacker.escalate.audit.su.login
intranet_server	2022-01-24 04:38:06 UTC	USER_CMD	None	pta1	None	None	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events, att
intranet_server	2022-01-24 04:38:06 UTC	CRED_REFR	PAMsetcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_START	PAMsession_open	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	USER_END	PAMsession_close	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events
intranet_server	2022-01-24 04:38:06 UTC	CRED_DISP	PAMsetcred	root	/usr/bin/sudo	/dev/pts/1	escalate, escalated_command, escalated_sudo_command	attacker.escalate.audit.sudo.command.events

View returned 9 rows: burst 1 = su from www.data to jhall, burst 2 = sudo cat /etc/shadow as jhall.

PANELS 2 TO 3 · CUMULATIVE FOOTHOLD & SU-THEN-SUDO TIMELINE

4. Where else does the attack leave traces

5. Detection rules ranked by lines triggered

PANELS 4 TO 5 · CROSS-HOST TRACES & RULE RANK

6. Audit event types touched by the chain

7. Busiest audit day on intranet_server

PANEL 7 · BUSIEST AUDIT DAY · ATTACK DAY = 3/4

8. Index improvement on the incident-response query

PANEL 8 · INDEX IMPROVEMENT · EXPLAIN SUMMARY

Attack day on intranet_server ranks 3 of 4 days by raw audit volume. *Volume-only detection misses it; the dashboard surfaces the chain anyway.*

Three findings the schema lets us defend.

01

Volume is not coverage.

Privilege escalation has only 9 audit events, yet it is the most consequential phase of the attack.

02

Volume-only detection misses the attack.

Attack day ranks 3 of 4 by raw audit volume on intranet_server. The chain hides inside baseline noise.

03

Relational design enables cross-source reconstruction.

Host inventory + audit events + labels + rules in one query is what made the 9-event chain readable.

DELIVERABLES	
GITHUB REPO	22-table 3NF schema · Alembic migrations · 23 SQL queries · Streamlit dashboard · README run steps.
TECHNICAL REPORT	10-section PDF; full SQL portfolio in section 6.
DASHBOARD DEMO	Live Streamlit app with screen-recording fallback.
WHAT HAPPENED NEXT	Exfiltration on internal_share is documented in the report appendix.

FLOOR OPEN

Questions?

Naman Sudan River

Roseveare-Hunt

Ishaan Shetty